Holiday Retail Planner 2026

Forecasting the U.S. holiday shopping season - and whether AI chatbots can do it

UCLA Anderson · Time Series Analysis and Forecasting (Dr. William Yu) · Group project, October 2026
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Live dashboard: https://holiday-retail-planner.streamlit.app **Code:** https://github.com/RachelLiu13/holiday-retail-forecast

1. Objective and motivation

November and December bring about **18% of U.S. retail sales** in two months, and over 22% for clothing stores. Retailers commit to holiday inventory in late summer, so the forecast they make in August decides whether shelves are empty or overstocked in December. Large chains have forecasting teams; small retailers mostly order *'same as last year'*. We ask: (1) how much will Americans spend this holiday season, by category; (2) can forecasting models beat 'same as last year', and which one; (3) can large language models (Claude, GPT) forecast as well as statistical models, or do they just remember published data; (4) does consumer sentiment improve the forecast; and (5) how can a small shop turn a national forecast into an order? The product is an interactive dashboard that answers all five.

2. Data

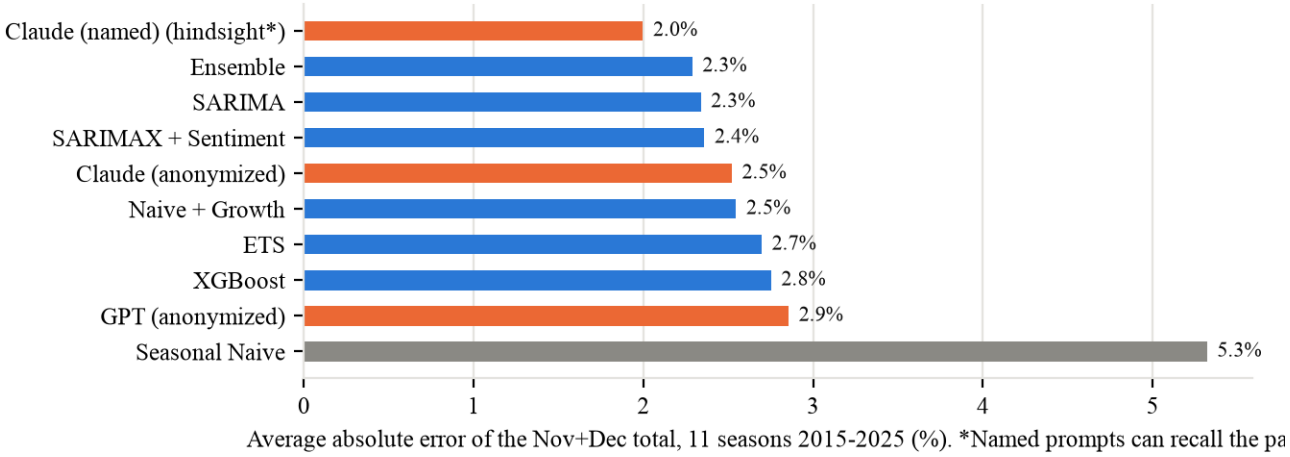
U.S. Census Bureau **Monthly Retail Trade**, not seasonally adjusted, $ millions, via FRED, January 1992 to August 2026 (total retail to September 2026). Seven series: total retail excluding food services, clothing, electronics & appliances, sporting goods/hobby/book/music, general merchandise, electronic shopping & mail-order (e-commerce), and grocery as a control with almost no holiday peak. Customer-side indicator: University of Michigan **Consumer Sentiment**. Checks found no missing months or duplicates. The March-June 2020 lockdown months are treated as outliers in model training, replaced with their pre-pandemic trajectory using only data available at the time.

3. Methods and validation

Model	How it works
Seasonal Naive (baseline)	This December = last December - the retailer's habit
Naive + Growth	Last year's month × recent year-over-year growth
ETS	Damped-trend exponential smoothing with monthly seasonality, on log sales
SARIMA	Seasonal ARIMA on log sales; order chosen by AIC from 16 candidates on training data only
SARIMAX + Sentiment	SARIMA + consumer sentiment lagged 4 months (always known at forecast time)
XGBoost	One gradient-boosted model pooled across categories; predicts year-over-year growth
Ensemble (recommended)	Average of SARIMA, ETS and XGBoost in log space
Claude Opus 5.5 / GPT	LLM given the same monthly history in a prompt, asked for the next 4 months (JSON)

Rolling-origin backtest. For each of 11 holiday seasons (2015-2025), every model stands at the end of August, is fit only on data available then, and forecasts September-December - the same origin, horizon and information for all models. The main metric is the absolute % error of the **November+December total**, the number a retailer orders against; we also report monthly MAPE, MASE and 80%-interval coverage. MAPE is suitable because sales are never near zero. **Memorization control for LLMs:** an LLM may have read the actual Census figures during training, so the main LLM test is *anonymized* (an index with first month = 100, labelled only by calendar month - no category, no year). For Claude we also ran *named* prompts with real dates and dollars; a large gap between the two would indicate recall rather than skill. Every LLM prompt and response is saved in the repository.

4. Results

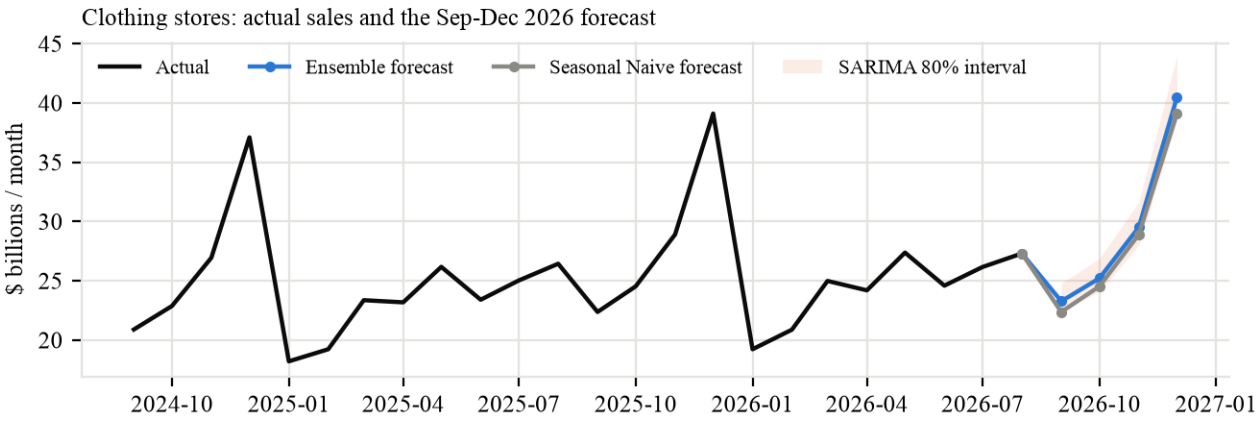


- **Forecasting beats habit.** 'Same as last year' missed the holiday total by 5.3% on average; the ensemble missed by **2.3%** and beat the baseline in 77% of category-seasons. Excluding the COVID seasons the gap is 3.8% vs 2.2%.
- **Model choice matters less than having a model.** SARIMA, the ensemble and 'last year × growth' are within a few tenths of a point; with 11 seasons these differences are not decisive. Electronics, hobby and e-commerce are the hardest categories; total retail and grocery the easiest.
- **Sentiment does not help** (error change +0.02 points): shoppers' mood has been weak while spending kept growing.
- **The intervals are honest:** SARIMA's 80% intervals contained 83% of actual values, so we use the backtest error spread for the planner's scenarios.
- **LLMs - hindsight, not foresight.** Told the category and the year, Claude looked like the best forecaster of all (2.0% average error). With them hidden, its error rose to 2.5%. The named prompt won in 65% of 77 category-seasons ($p = 0.002$, statistically significant), and the gap was largest in 2020-2022 (+1.2 points vs +0.5 overall) - exactly when knowing what really happened (the pandemic, inflation) helps most. That is hindsight, not forecasting, so only anonymized scores measure skill.
- **Fairly tested, LLMs are good but not better.** On anonymized data, the chatbots missed the holiday total by Claude 2.5%, GPT 2.9%, vs 2.3% for our ensemble and 5.3% for 'same as last year'; the ensemble was better than Claude in 57% of cases ($p = 0.08$, not statistically significant); the ensemble was better than GPT in 61% of cases ($p = 0.004$, statistically significant).

2026 forecast (ensemble, data through August 2026). Total holiday retail sales of $1,383 billion, +5.2% over 2025. E-commerce keeps gaining share (17.4% of retail in Jan-Aug 2026 vs 16.9% a year earlier).

Category	Nov+Dec 2025 actual	Nov+Dec 2026 forecast	Growth
Total Retail (ex. food services)	$1,314.7B	$1,383.1B	+5.2%
Clothing & Accessories Stores	$68.0B	$70.0B	+2.9%
Electronics & Appliance Stores	$18.2B	$19.3B	+5.8%
Sporting Goods, Hobby, Book & Music Stores	$20.8B	$22.4B	+7.7%
General Merchandise Stores	$179.1B	$185.7B	+3.7%
E-commerce (Electronic Shopping & Mail-Order)	$258.5B	$283.3B	+9.6%
Grocery Stores (control: no holiday peak)	$175.7B	$177.7B	+1.1%

Early reality check: September 2026 total retail ($635.1B) was published after our forecast origin; the ensemble's forecast was off by -1.7%.



5. How to use the dashboard

- **Story** - the headline results. **Market** - history, the size of each category's December spike, e-commerce share.
- **2026 Forecast** - choose a category and models; SARIMA's 80% interval is shown as a band.
- **Model Battle** - backtest scores with and without COVID seasons, errors by category, a replay of any past season, and the LLM memorization test.
- **Holiday Planner** - a small retailer enters last season's Nov+Dec sales, stock on hand and on order, a service level, and how volatile the store is. The tool returns expected sales (own history × national category growth), a safety buffer ($z \times \sigma \times$ expected, where σ combines our past national forecast error and the store's own volatility), the recommended order, and conservative/optimistic scenarios.
- **AI Advisor** - questions in plain English (e.g. 'Why this order?', 'Which model is best?'). Claude answers only from the computed results it is given and says so when information is missing; it never produces forecasts or ranges itself.

6. Limitations

- National category totals, not a specific store; the planner scales the user's own history by the national trend.
- FRED holds today's revised data; forecasters in past years saw earlier, unrevised figures.
- Eleven seasons is a small sample. No inventory-cost data, so we make no claim about money saved.

7. AI tools used

- **Claude Code** (Anthropic) wrote most of the data pipeline, models, backtest, dashboard and first drafts of this description and the slides, from the team's research questions and design decisions; the team reviewed the code and checked results against the raw data.
- **Claude Opus 5.5** and **OpenAI GPT** are forecasters in the model comparison (course Option A).
- **Claude Opus 5.5** powers the dashboard's AI Advisor (course Option B).